



Slump Test App.



Aimil Ltd.

Instrumentation & Technologies

Version

1.0



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AIM – 334

SLUMP TEST APPARATUS

Operating Instructions



Operating Instructions Slump Test Apparatus

Introduction

The Slump Test Apparatus is used for the determination of consistency of concrete where the maximum size of the aggregate does not exceed 38mm and meets the basic requirements of IS:7320, BS:1881, ASTM C-143 and AASHO T-119. The test can be carried out either in the laboratory or during the progress of work in the field.

Description

The equipment is illustrated in the General Assembly Drawing. The numbers given against the components of the equipment in the description below pertain this General Assembly Drawing.

It consists of :

1. SLUMP CONE

The Slump Cone (2) is in the form of a frustum of a cone with two lifting handles (3) to facilitate the lifting of the cone from the moulded concrete test specimen. It has two foot pieces (5) to facilitate locking with the base plate.

2. BASE PLATE

The base plate (1) has four cleats. One of the cleats has a hole through which the tamping rod (7) passes. There are two clamps (4) with nuts to clamp the slump cone to the base. A swivel arm (6) is fixed to the base. It acts as handle to carry the entire assembly and in the raised position it also acts as datum from which point the slump of concrete is measured.

3. TAMPING ROD

One end of the steel tamping rod (7) has graduations at 1 cm intervals from 0 to 20 cm while the other end is rounded.



Sampling

a. FIELD

Take truly representative samples from the batch of the composite sample obtained from the freshly mixed concrete making it 0.02 cm³ in volume. When the continuous mixers are used, a batch shall be regarded as the discharge during one minute. In case of concrete containing aggregates of a maximum size of more than 38mm, concrete shall be wet sieved through a 38mm screen to exclude aggregate particles bigger than 38mm.

i. *From Mixers*

Take at least three representative and equal samples of 0.02 cm³ in volume from a batch during its discharge from a mixer, by passing a clean and dry receptacle with retaining sides across the stream of concrete. The receptacle is to be constructed of non-absorbent material, preferably of metal and should be such that the sample retained is not segregated.

If only three sample intervals are desired, take them at about the time when one quarter, one half and three quarters of the concrete have been discharged from the mixer and if more than three are required, take them at correspondingly shorter but equally spaced intervals.

ii. *From Concrete at the time and place of Deposition*

While a batch of concrete is being discharged or immediately after it has been discharged, collect the sample from not less than five well distributed positions, avoiding the edge of the mass where segregation might have occurred.

iii. *Mixing the Composite Sample*

Mix the sample obtained by either of the methods described above on a non-absorbent base with a shovel or any other suitable implement in such a manner as to ensure uniformity.

b. LABORATORY

As accurate control of the quantities of material and test condition is possible, this test may preferably be carried out in the laboratory. Take the representative samples of all the concrete materials. Test samples of concrete shall be taken from larger lots by quartering.



Mixing Concrete

Mix all concrete materials either by hand or in a laboratory batch mixer, in such a manner as to avoid the loss of water or of other materials.

i. Machine Mixing

Where the mixing drum is power loaded, introduce water before the solid materials are added. Load the skip with about one-half of the coarse aggregates, then with the cement and finally with the remaining coarse aggregates on top (if all the aggregates are used, load the skip first with one half of the aggregates then with the cement and finally with the remaining aggregates on top). If the mixing drum is hand loaded, it shall be charged with dry materials in a similar manner but the water shall be added immediately before the rotation of the drum is started. The period of mixing shall be not less than two minutes after all the materials are in the drum and mixing shall continue until the resulting concrete is uniform in appearance. When using pan mixer, the concrete shall be heaped together before sampling.

ii. Hand Mixer

Mix the concrete batch on a non-absorbent platform with a shovel, trowel or similar suitable implement following the procedure given below :

- a. Mix the dry cement and dry fine aggregate until the mixture is thoroughly blended and uniform in colour.
- b. Add the coarse aggregate and mix with cement and fine aggregate until the coarse aggregate is uniformly distributed throughout the batch.
- c. Add water and mix the entire batch until the concrete appears to be homogeneous.

Protect the same from the weather.



Setting up and Test Procedure

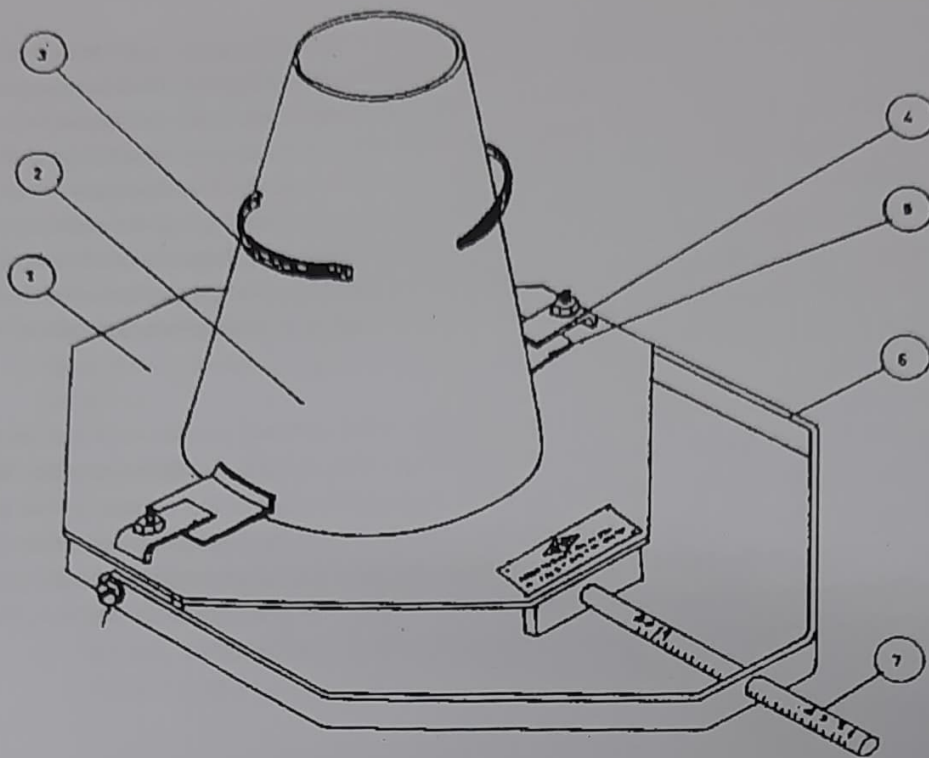
Clean the internal surface of the mould and make sure that the surface is free from superfluous moisture and any set concrete. Apply a thin coat of grease to the internal surface of the slump cone and the base plate. Position the slump cone on the base plate which is positioned on a level surface.

Clamp the slump cone to the base. Fill it with concrete in four equal layers, each layer being given 25 well distributed strokes of the rounded end of the tamping rod. While tamping the first layer, the rounded end of the tamping rod shall reach the bottom and for the rest of the layers it shall reach the preceding layer. Level off the surface and unclamp the slump cone immediately and slowly lift and remove the cone vertically. The concrete will now subside. Raise the swivel arm to its vertical position and touch the highest point on the specimen with the tamping rod passing through the swivel arm. Take the reading on the tamping rod which gives the slump in centimeters. This test shall be carried out within a period of 22 minutes after sampling and the place of testing should be free from vibrations or from any type of shock.

Any slump specimen which collapses or shears off laterally will give an incorrect result, and if this occurs, the test shall be repeated with another sample. If with the repeated test the same thing happens, record it as such. Some indication of the cohesiveness and workability of the mix can be obtained in this manner. After the slump measurement has been completed, the sides of the concrete should be tamped gently with the tamping rod. A well proportioned concrete which has an appreciable slump will gradually slump further, but if the mix has been badly proportioned, it is likely to fall apart.

Maintenance

After completing the test, the inner surface of the slump cone should be thoroughly cleaned and greased. Any concrete left sticking to the inner surface would adversely effect further testing.



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S.NO.	DESCRIPTION
1	Base Plate
2	Slump Cone
3.	Lifting Handle
4.	Clamp



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S.NO.	DESCRIPTION
5.	Foot Piece
6.	Swivel Arm
7.	Tamping Rod